DATA INTEGRATION WITH ETL

STRUCTURE

Why



Learning basics features of ETL. Creation of a routine in all its component. Running routines and jobs

How

- Live class, each topic delivered by the pattern
  - theory
  - demo
  - Workshop
- Material
  - trainer
    - Slides (with notes)
    - VM with training environment
    - This map
      - **Legend**
      - Main message
      - Point of Attention
      - Tips & Tricks
      - Theory/Demo
      - Workshop
  - attendees
    - QRCs (coming soon)
    - Slides (printed)
    - Workshops (printed and html)
    - VM with training environment
- Assumption: Version used 5.3.11
- Out of scope
  - Advanced features of ETL
  - Advanced use of transformations
  - Advanced use of SQL on routines

Who

- Audience
  - Partners: Basic implementators (no project design capabilities)
  - Customers Administrators
  - Tagetik attendees
- Pre-requisites
  - Data model
  - Output reporting
  - Data collection
  - SQL skills
  - Novice level

INTRODUCTION

- Why use ETL
  - Mention that ETL can be used to upload data and metadata
  - Recap of all possible way to load data in Tagetik
  - Explain when ETL is the right solution

TERMINOLOGY

- ETL terminology
  - Explain the meaning of achronym E.T.L.
  - Explain the meaning of Domain, Routine and Job following slides
  - Show in the application where all this component are defined

ARCHITECTURE

- ETL Architecture
  - Show the slides describing the ETL architecture and all possible sources and targets
  - Decribe the main attribute of ETL commenting the related slide

CONFIGURATION

- Staging table fields
  - Staging table fields are the core of ETL follow the slides and explain how many staging table field are available for each domain.
  - Focus on the fact that STF are defined for each domain not for a single routine
  - Open on SQL Tagetik tables MAP_DATI_ACQUISITI and MAP_DATI TRASFORMATI and mention the relationship between STF and the columns of these tables
  - Theory/Demo
    - Create three STF focusing on all attributes
    - Mention the possible code values and that it represent the number of the STF
    - Mention why we could need of Achronym to refer a field
    - Mention all the possible types and the meaning of Identity
    - Explain how to assign a dictionary to a field and all the different possibility of dictionary
- Parameters
  - Explain the use of parameters in the 3 phases of ETL
  - Theory/Demo
    - Explain the different type of parameter
    - Include also the Data Intersection explanation
  - Workshop

SETUP EXTRACTION

- Extraction
  - Theory/Demo
    - Mention the different kind of data sources available
    - Remind that one routine can hold many data sources
    - Show and describe attribute of different datasources in tabs files and databases (e.g. while extracting data from .xlsx file print Area definition is mandatory)
  - Workshop

TRANSFORMATIONS

- Explain that most of transformations are defined ad-hoc in the routines. Lookup table are an exception that is defined at level of domain. It is usual that many routines of the same domain should use the lookup table defined
- Lookup tables
  - Some of other transformations will be debated later on in routine definition
  - Theory/Demo
    - Used to map one or more input value with one or more output value
    - Maintenance done by user normally by an ETL that fills table MAP_REGOLA_TAB_ELEMENTO
    - Show slides with examples and mention use of wildcard characters
  - Workshop

SETUP LOADING

- Loading
  - Theory/Demo
    - Mention the different data targets available
    - Remind that one routine can load data contemporarily on different data target
    - Tagetik datamodel pdf can be downloaded from web application. Useful to know name of tables and columns
    - Grouping functions on different fields to be mapped
    - List of available grouping functions can be shown in tooltip of Aggregate function textbox.
    - Mention the meaning of Update type "by value"
  - Workshop
- Recap

ETL ROUTINE

- ETL Routine creation
  - Theory/demo
    - Routine creation is the container of all the Domain component defined so far in the training.
    - In a routine are defined all the information of Extraction, Transformation and Loading. The result of the definition is a data processing that can be run as single or that can be included in a Job with other routines
    - Explain routine attributes (e.g. how to store data on staging tables)
    - Focus on data processing frame with its useful options
    - Show how to include the parameter in the different phases
    - In transformation phase comment the transformations included in the example but even the other important ones as Year-to-date calculation, Verticalization and what you retain useful to trainees. You can provide also numeric examples
    - Comment all the transformations used to calculate the amount, to retrieve the Currency from Azienda table, to apply a fixed value to a dimension (category)
    - While introducing loading phase focus on different loading policy and in which situation one is to be preferred to the others
    - Explain how to avoid loading zero or null elements using field ETL Filter
  - Workshop

ETL JOB

- ETL job
  - Introduce concept of ETL job and explain the features showing slide
  - Theory/Demo
    - Create the jobs mentioning the details (Routines, Parameter, Extractions)
    - Focus on the fact that a Job can be scheduled (for example overnight every first day of month)
    - Mention all the way to run an ETL Routine and an ETL Job
  - Workshop

TROBLESHOOTING & TOOLS

- Troubleshooting
  - Logs are shown while running routine but also stored
  - Functional errors (e.g. entity code doesn't exists)
  - Technical errors (SQL errors)
  - Staging table viewable in Tagetik or in SQL. Used to check data flow
  - Use of drill down and drill through
- ETL tools
  - Mention the common ETL tools: Import/Export, Delete ETL, Copy ETL

Questions and Answers

Morning

Afternoon